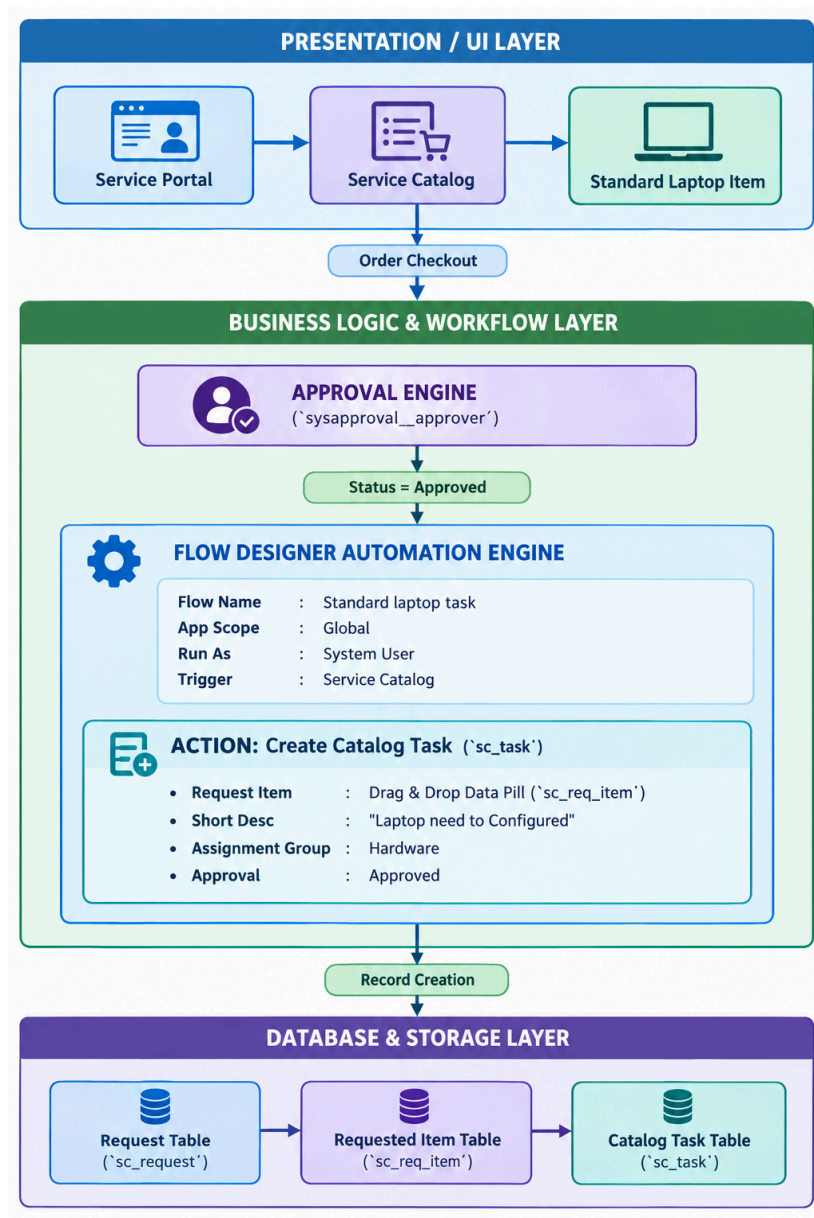


Project Design Phase Solution Architecture

Date	28 October 2026
Team ID	SWTID-2026-6819
Team Leader	Hazna Mariyam. H
Team Members	Anuskha. K Theras Jasmine. M
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

1. System Architecture Diagram

The system architecture follows a 3-tier cloud model hosted natively on the ServiceNow platform, separating user interactions, automation workflows, and core database entities.



2. Technical Component Specifications

2.1 Service Catalog Binding

- **Catalog Item:** Standard Laptop
- **Category:** Hardware
- **Process Engine Mapping:** Bound to **Standard laptop task** flow under **Maintain Items** -> **Process Engine** tab.

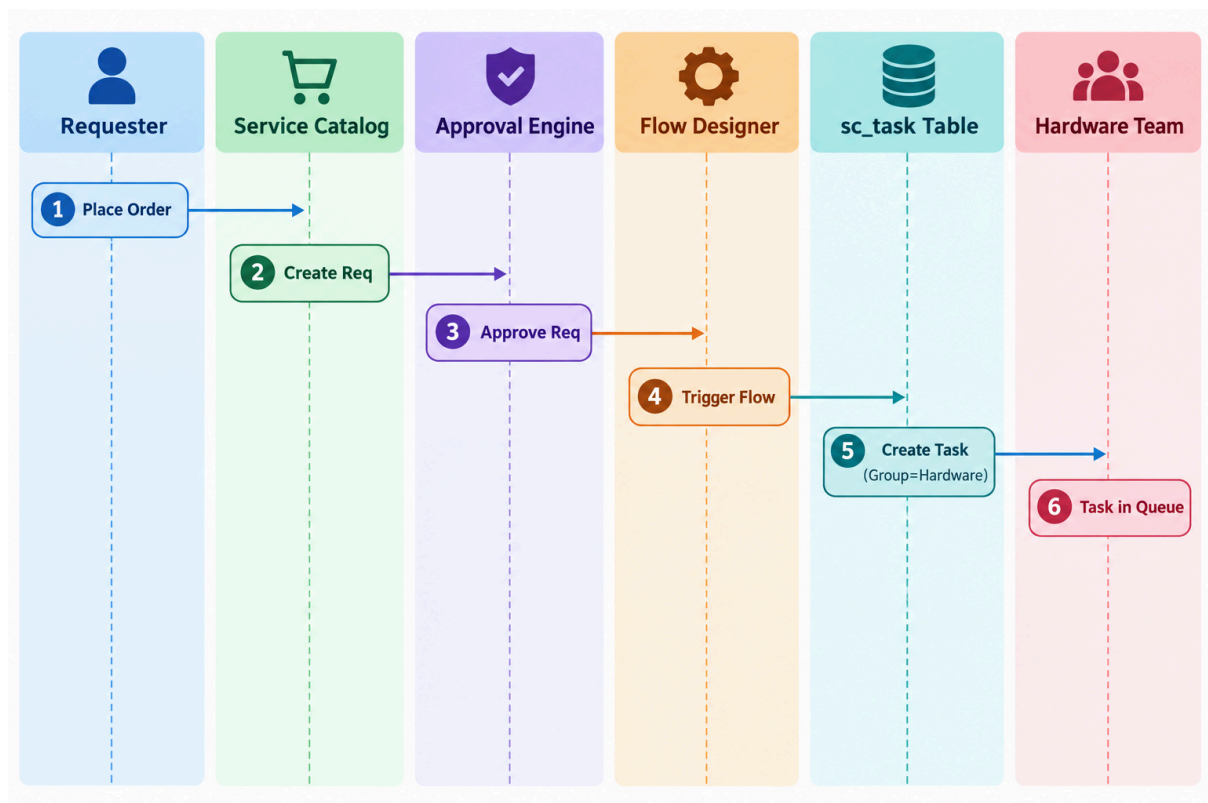
2.2 Flow Designer Configuration Details

- **Flow Name:** Standard laptop task
- **Application Scope:** Global
- **Run As:** System User (Bypasses requester-level ACL restrictions)
- **Trigger Type:** Service Catalog

2.3 Action Properties (Create Catalog Task)

- **Target Table:** Catalog Task (sc_task)
- **Request Item Binding:** Trigger -> Requested Item Record
- **Short Description:** "Laptop need to Configured"
- **Description:** "Laptop need to Configured"
- **Assignment Group:** Hardware
- **Approval State:** Approved

3. Data Flow & Interface Sequence Diagram



4. Execution & Security Controls

1. Transaction Isolation: Flow actions execute within standard ServiceNow system transaction boundaries.
2. Access Control Lists (ACLs): Execution under System User ensures smooth record insertion in `sc_task` even if the requester lacks direct write privileges to fulfillment tables.
3. Auditability: Every flow execution generates an immutable log in Flow Designer -> Execution History, capturing timestamps, data pills, and execution status.